

Ninja Code – Simple Hinglish Summary

Source: <https://javascript.info/ninja-code>

Main baat

Ye chapter **sarcasm (vyang)** me likha gaya hai. Isme "ninja programmers" ki wo **bure tricks** batayi gayi hain jinse code padhna aur maintain karna **mushkil** ho jata hai. Asli maqsad ye hai ki tum **pehchano ki kya NAHI karna chahiye**.

Har "advice" ko ulta padho: jo yahan "achha" bataya gaya hai, wo asal me **galat practice** hai.

Author ke shabdon me: ye sab advice **real code** se li gayi hain, kabhi kabhi experienced developers ne bhi likhi hain.

Ninja tricks (aur unka sahi ulta)

1. Code ko bahut chhota likho ("Brevity")

Ninja way: Code jitna ho sake chhota likho, language ke subtle features use karke apni hoshiyari dikhao.

```
javascript

// ek famous library se
i = i ? i < 0 ? Math.max(0, len + i) : i : 0;
```

Problem: Dusra developer is line ko samajhne me bahut time lagayega. **Sahi tarika:** Code **readable** likho, chhota nahi.

2. Ek-akshar wale variables

Ninja way: Har jagah (a), (b), (c) jaise naam use karo. Loop counter me (i) ki jagah (x) ya (y) use karo, khaaskar jab loop body 1-2 page lambi ho.

Problem: Editor ke search me nahi milta, aur naam se kuch pata nahi chalta. **Sahi tarika:** **Descriptive naam** rakho.

3. Abbreviations

Ninja way: (list) → (lst), (userAgent) → (ua), (browser) → (brsr).

Sahi tarika: Naam ko chhota mat karo, poora aur saaf likho.

4. Bahut abstract naam

Ninja way:

- Sabse ideal naam (data) hai, use har jagah lagao. (data) le liya to (value) use karo.
- Variable ko uske **type** ke naam se bulao: (str), (num).

- Naam khatam ho jaye to number jodo: `data1`, `item2`, `elem5`.

Problem: Naam sirf ye batata hai ki andar kya type hai, ye nahi ki **kya cheez** hai. **Sahi tarika:** Naam batae ki variable **kya store karta hai** (jaise `userName`, `shoppingCart`).

5. Attention test (milte-julte naam)

Ninja way: `date` aur `data` jaise milte-julte naam mix karo.

Problem: Jaldi padhna namumkin, aur typo hone par bahut der tak atke rahoge.

6. Smart synonyms

Ninja way: Ek hi tarah ke kaam ke liye alag alag prefixes use karo (`displayMessage`, `showName`, `renderTitle`, `paintText`). Aur **alag kaam** wale functions ke liye **same prefix** use karo: `printPage` (printer par), `printText` (screen par), `printMessage` (naye window me).

Problem: Padhne wala confuse ho jata hai ki kaunsa function kya karta hai. **Sahi tarika:** Ek hi kaam ke liye **ek hi prefix**, aur alag kaam ke liye alag.

7. Naam reuse karo

Ninja way: Naya variable tabhi banao jab bilkul zaruri ho. Purane me hi nayi values likhte raho. Function ke beech me value ko chupke se replace kar do:

```
javascript

function ninjaFunction(elem) {
  // 20 lines of code working with elem

  elem = clone(elem);

  // 20 more lines, ab clone par kaam ho raha hai!
}
```

Problem: Pata hi nahi chalta ki variable me abhi kya hai. **Sahi tarika:** **Alag values ke liye alag variables.** Extra variable achha hota hai, bura nahi.

8. Underscores mazaak ke liye

Ninja way: Variables ke pehle `_` ya `_` lagao, bina kisi matlab ke, ya alag jagah alag matlab ke saath.

Problem: Code lamba aur kam readable ho jata hai.

9. "Super", "mega", "nice" jaise naam

Ninja way: `superElement`, `megaFrame`, `niceItem` jaise naam rakho.

Problem: Ye kuch details nahi dete. Padhne wala matlab dhundhta rehta hai.

10. Bahar ke variable ko overlap karo

Ninja way: Function ke andar aur bahar same naam use karo:

```
javascript

let user = authenticateUser();

function render() {
  let user = anotherValue();
  ...
  ...many lines...
  ... // <-- programmer yahan user use karna chahta hai...
}
```

Problem: Programmer ko pata hi nahi chalta ki local `user` ne bahar wale ko **shadow** kar diya hai.

11. Side-effects har jagah

Ninja way: `isReady()`, `checkPermission()`, `findTags()` jaise functions me "useful" extra action daal do jo bahar ka kuch badal de. Ya `true/false` ki jagah ek complex object return karo.

Problem: Sab maante hain ki `is...`, `check...`, `find...` functions kuch badalte nahi. Aise me `if (checkPermission(...))` kaam nahi karega. **Sahi tarika:** Function jo naam kehta hai wahi kare, usse zyada nahi.

12. Powerful functions

Ninja way: Function ko naam se zyada kaam karne do. Jaise `validateEmail(email)` email check karne ke saath error message bhi dikhaye aur dobara email maange.

Problem: Kisi ko sirf email check karna ho to ye function uske kaam ka nahi, kyunki ye **reuse se bachata hai**. **Sahi tarika:** Ek function, ek kaam.

Summary

Ye saari "advices" asli code se aayi hain. Inhe follow karoge to:

- Kuch follow karo to code **surprises se bhar jayega**.
- Bahut saari follow karo to code **sirf tumhara** ban jayega, koi badalna nahi chahega.
- Sab follow karo to code **naye developers ke liye ek seekh** ban jayega.

Quick Table: Ninja vs Sahi tarika

Ninja trick (galat)	Sahi tarika
Bahut chhota, "smart" code	Readable code
<code>a</code> , <code>b</code> , <code>x</code> jaise naam	Descriptive naam
<code>lst</code> , <code>ua</code> , <code>brsr</code>	Poore aur saaf naam
<code>data</code> , <code>value</code> , <code>str</code>	Batao variable kya store karta hai
<code>date</code> aur <code>data</code> mix	Milte-julte naam mat rakho
Same kaam, alag prefixes	Same kaam, same prefix
Purane variable me nayi value	Naya variable banao
Bekaar <code>_</code> , <code>_</code>	Sirf jahan matlab ho
<code>superX</code> , <code>megaY</code>	Specific naam
Andar-bahar same naam	Alag naam (shadowing se bacho)
<code>check...</code> me side-effect	Function sirf naam ka kaam kare
Ek function me kai kaam	Ek function, ek kaam